Structural Mode N in CPTG

Measuring Gravitational Organization from Galaxies to Black Holes

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Abstract

Curvature Polarization Transport Gravity (CPTG) is a geometric theory of gravitation in which ordinary matter provides the physical source while curvature polarization and curvature transport describe the organized gravitational response. Within this framework, Structural Mode N is a dimensionless post-solution measurement of how that completed gravitational state is internally organized. Defined generally by

$$N = \frac{R}{\lambda},$$

it compares the structural extent R of an object with a geometry-appropriate organization length λ. Structural Mode is therefore not a fit parameter, a prediction assigned before solution, or a measure of gravitational strength. It is a structural fingerprint extracted from the solved state itself.

The same measurement principle applies across distinct gravitational regimes while the organization operator follows the geometry of the object. In resolved galaxies, N measures the radial organization of the solved gravitational field and provides a morphology-blind structural coordinate that can recover organization not supplied by conventional classification. The Curvature Structure Mode Index (CSMI) provides a downstream catalog representation of solved N populations; its descriptive bands may expand, subdivide, or otherwise evolve as larger and more diverse samples are analyzed. In pressure-supported galaxy clusters, Structural Mode is reconstructed from the solved spherical field and remains specific to each independent structural reconstruction rather than being averaged across them.

For finite rotating black holes, Structural Mode acquires an additional capability. The compact realization assigns an internal organization shell whose location is fixed by the solved mass–spin geometry. At that shell, the finite geometry supplies definite time-dilation and frame-dragging readouts. Structural Mode therefore identifies not only where a gravitational configuration is organized, but, in the compact regime, where the local structure of time and rotation can be measured within that organization. Across galaxies, galaxy clusters, and black holes, N provides a common structural language for comparing gravitational organization without replacing the regime-specific physics that generates each solved state.

1 Introduction

Astronomical objects are usually described through what can be observed most directly. Galaxies are given morphology labels from their luminous appearance, clusters are described through gas and pressure structure, and black holes are characterized through mass, spin, and the geometry inferred from their exterior state. Those descriptions are useful, but they do not necessarily answer the same physical question: *how is the gravitational state itself organized?*

Curvature Polarization Transport Gravity (CPTG) provides a way to ask that question only after the gravitational problem has been solved. The theory keeps ordinary matter as the material source of gravity and adds geometric response terms that describe how curvature is polarized by structure and how that organized response is transported through the system [1]. The object is therefore not classified first and then forced into a gravitational template. Its measured source, physical state, geometry, and boundary conditions are used to obtain the CPTG solution; structural information is read from that solution afterward.

Structural Mode N is that post-solution readout. This paper establishes the broader CPTG structural hierarchy now realized across resolved galaxies, pressure-supported galaxy clusters, and finite rotating black holes. Across these regimes, each gravitating system carries its own structural acceleration scale or equivalent; the appropriate source–polarization–transport response is solved in the geometry of that regime; and Structural Mode is measured only from the completed state.

The central distinction is between *solving gravity* and *describing the organization of the solved state*. Structural Mode does not tell CPTG which gravitational solution to choose. It does not determine the source strength, the galaxy morphology, the cluster dynamical class, or the black-hole spin. Instead, after the state has been fixed, it measures where the important structural changes are concentrated relative to that object's own solved domain.

For galaxies, this separation creates an unusually direct test. The SPARC catalog supplies resolved rotation curves and baryonic source components together with independent catalog information such as Hubble type, surface brightness, luminosity, size, H I mass, and flat rotation velocity [8]. The gravitational calculation uses the rotation state and baryonic source. Structural Mode is measured afterward. The catalog descriptors can then be consulted as non-input comparison coordinates. Agreement is informative because the catalog label was not used to produce N; disagreement can be even more informative if deeper observations reveal organization that the conventional label obscures.

The same logic can be carried to other geometries. A spherical cluster requires a spherical organization measure, not a disk-galaxy formula. A rotating black hole requires a relativistic compact measure, not a cluster shell integral. The quantity that survives across these regimes is the physical statement

$$N = \frac{R}{\lambda}, \tag{1}$$

where R is the outer structural extent of the declared solved state and λ is the geometry-appropriate organization length extracted from that state. The analysis therefore proceeds from the CPTG structural framework to three distinct physical regimes—galaxies, galaxy clusters, and black holes—before comparing the resulting Structural Mode measurements across scales.

2 CPTG Framework and Structural Coordinates

The framework used here separates quantities that construct the gravitational state from quantities read from that state afterward. Ordinary matter supplies the material source; an object-dependent structural acceleration scale or regime-equivalent participates in the CPTG response; Structural Mode is a downstream measurement of the completed state.

2.1 Source, polarization, and transport

CPTG is formulated as an extension of General Relativity rather than as a replacement for its geometric foundation [1]. At the parent-field level,

$$G_{\mu\nu} + \Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \tag{2}$$

$T_{\mu\nu}$ is the ordinary material stress–energy source. The additional terms are geometric responses. Curvature polarization describes how the gravitational response reorganizes according to the strength, gradients, and structure of the source-supported field. Curvature transport describes how that organized response is retained or redistributed through an extended or evolving system. When these additional responses become negligible, the solution approaches the ordinary General Relativistic response sourced by the same matter.

The common ordering relevant to this paper is compactly summarized by

$$\begin{aligned} \text{source} + \text{measured state} &\longrightarrow \text{structural acceleration} + \text{response coordinates} \\ &\longrightarrow \text{polarization} + \text{transport} \longrightarrow \text{solved state} \\ &\longrightarrow \text{Structural Mode and other post-solution readouts.} \end{aligned} \tag{3}$$

The structural acceleration scale therefore belongs to the solution itself, whereas Structural Mode belongs to the subsequent structural readout.

2.2 Structural acceleration $a_\star$ and regime equivalents

The structural acceleration is one of the principal state coordinates of CPTG. Every mass-bearing gravitating system carries an object-dependent acceleration scale associated with its physical structure. The symbol $a_\star$ is used where that scale appears explicitly in the reduction; in other regimes the same structural role can be carried by the natural

acceleration normalization of the solved geometry. It is therefore not a universal constant shared by all objects, and it is not restricted to galaxies. The coordinate realization changes with the physical regime while the structural role is retained [1, 6].

For the resolved galaxy and pressure-supported cluster reduction used in this paper, the structural acceleration is local:

$$a_\star(r_i) = g_{\rm sup}(r_i). \tag{4}$$

Each measured radius therefore carries its own $a_\star$. No object-wide average is substituted for the resolved profile. In a rotating galaxy, $g_{\rm sup} = V_{\rm obs}^2/r$. In a pressure-supported cluster, $g_{\rm sup}$ is reconstructed from the measured hydrostatic or Jeans support state.

For a stationary compact black hole, the same structural-acceleration principle is represented by the mass-scaled strong-field acceleration normalization. The retained exterior gravitational mass fixes

$$r_m \equiv r_g = \frac{GM_\infty}{c^2}, \qquad g_g = \frac{c^2}{r_m} = \frac{c^4}{GM_\infty}. \tag{5}$$

For the registered compact realization, r_m is the gravitational matching scale and is identical to the exterior gravitational radius r_g. The compact field can therefore be written in mass-scaled form, $x = r/r_m$ and $g = g_g U(x)$, while the dimensionless spin completes the stationary compact state. In this regime g_g is the strong-field acceleration equivalent of the structural scale represented explicitly by $a_\star$ in the resolved calculations. The notation changes because the coordinate realization changes; the underlying CPTG requirement that the gravitational response be tied to an object-specific structural acceleration scale does not.

In the resolved galaxy and cluster realizations, the explicit $a_\star$ coordinate enters the structure-dependent response rather than being reported as an auxiliary number. It weights the polarization contribution and appears in the nonlinear ratio between the solved acceleration and the local structural acceleration. The full resolved galaxy realization is written explicitly in Section 3.2; the cluster reduction uses the same local structural-acceleration role with a pressure-supported source state. In the compact regime the corresponding normalization is supplied by g_g together with spin.

Structural acceleration and Structural Mode are therefore complementary but not interchangeable. The former helps construct the CPTG state; the latter measures how the retained state is organized. A change in structural acceleration can alter the field or geometry from which N is subsequently measured, but N is not obtained by inserting an acceleration scale into a separate classification formula.

2.3 Structural Mode N

With the common coordinate already defined by Eq. (1), R denotes the structural extent of the declared realization and λ the geometry-appropriate location of structural change within the solved state. In the resolved galaxy and cluster calculations, R is the outer

retained radius of the particular reconstruction; in the compact realization the geometry is normalized by the matching scale r_m and evaluated on $0 \leq x \leq 1$. Lower N places the dominant structural change farther outward relative to the declared extent, whereas higher N places it farther inward.

The physical meaning of N is common across object classes, but the construction of λ follows the geometry of the solution. A rotating disk, a spherical pressure-supported cluster, and a finite rotating black hole therefore do not share one copied operator. Structural Mode is likewise not a fitted gravitational amplitude, branch selector, or prospective value assigned to an unsolved object. It is a post-solution fingerprint of the retained state. Independent reconstructions of the same object keep independent N measurements rather than being averaged into a synthetic value.

Table 1 summarizes the three realizations used in this paper.

Regime	Structural coordinate	Solved state	Mode realization
Galaxies	Local $a_\star(r) = g_{\mathrm{sup}}(r)$ from rotational support	Resolved radial CPTG acceleration field	Outer-field weighted centroid $\lambda_{N,\mathrm{gal}}$ on $R/5 \leq r \leq R$; $N_{\mathrm{gal}} = R/\lambda_{N,\mathrm{gal}}$
Galaxy clusters	Local $a_\star(r) = g_{\mathrm{sup}}(r)$ from pressure support	Solved spherical pressure-supported CPTG field	Three-dimensional shell-weighted centroid $\lambda_{N,C}$ on the retained radial domain; $N_C = R/\lambda_{N,C}$
Black holes	Strong-field normalization $g_g = c^4/(GM_\infty)$, with spin completing the compact state	Finite Kerr-compatible mass–spin geometry	Equal-measure rotating coordinate ξ with compact structural weight $w(x)$; $\lambda_{N,\mathrm{BH}}$ gives the organization length and $N_{\mathrm{BH}} = r_m/\lambda_{N,\mathrm{BH}}$

Table 1: Cross-regime Structural Mode architecture. Structural acceleration participates in forming the solved state; Structural Mode is then measured with the geometry-appropriate organization operator.

2.4 Curvature Structure Mode Index

A continuous Structural Mode is the primary measurement. The Curvature Structure Mode Index (CSMI) is a downstream catalog layer that assigns descriptive names to recurring intervals of the solved N distribution. CSMI does not create N, does not alter the solved field, and does not feed a class label back into the gravitational calculation. Its named bands are empirical catalog structure, not immutable constants of CPTG. The continuous N measurement has priority over any particular catalog partition.

For galaxies, the current CSMI catalog is

Structural Mode N	CSMI label
$N \leq 1.45$	Dwarf Irregular
$1.45 < N \leq 1.75$	Magellanic Irregular
$1.75 < N \leq 1.95$	LSB Dwarf Disk
$1.95 < N \leq 2.15$	Transition Dwarf
$2.15 < N \leq 2.35$	LSB Spiral
$2.35 < N \leq 2.55$	Very Late Spiral
$2.55 < N \leq 2.80$	Late Spiral
$2.80 < N \leq 3.05$	Intermediate Spiral
$3.05 < N \leq 3.30$	Early Spiral
$3.30 < N \leq 3.55$	Bulged Spiral
$3.55 < N \leq 3.72$	Lenticular/Early Disk
$N > 3.72$	High-Mode Outlier

Table 2: Current galaxy CSMI catalog. The continuous Structural Mode is measured first and the descriptive structural label is assigned afterward. The listed intervals are the present population-based partition, not permanently fixed boundaries of CPTG.

The numerical intervals in Table 2 are the registered downstream galaxy catalog partition carried into the present analysis. They are applied only after the continuous N value has been measured and are not re-fit to the current Hubble-type comparison. Object-level assignment uses the solved N value alone: galaxy name, catalog morphology, and the other withheld metadata do not enter the field solution, the calculation of N, or the interval lookup. The descriptive names summarize the structural character associated with the ordered mode intervals; they do not redefine the continuous measurement as a visual Hubble classifier. The present boundaries are therefore catalog coordinates rather than fundamental constants of CPTG. As larger and more diverse solved populations become available, future CSMI catalogs may expand, subdivide, merge, refine numerical boundaries, or adopt more informative names while leaving the underlying continuous N measurements unchanged.

CSMI is not intrinsically restricted to galaxies. A cluster, compact-object, or other object-class CSMI can be developed if its own solved N population exhibits reproducible structural bands; its thresholds and names must be derived from that population rather than copied from the galaxy catalog.

3 Galaxies

3.1 Observational inputs and validation firewall

The galaxy analysis provides the largest resolved sample in which the separation between gravitational inputs and structural outputs can be tested directly. The present paper uses the primary SPARC science sample drawn from the resolved rotation curves, baryonic mass-model components, and common metadata catalog of Lelli et al. [8]. The retained

science sample is defined by

$$Q \neq 3, \qquad i \geq 30^{\circ}, \tag{6}$$

where $Q = 3$ is the lowest SPARC quality flag and i is inclination. This leaves 153 galaxies containing 3168 resolved rotation-curve points.

The important methodological firewall is shown in Table 3. The left column contains quantities allowed to participate in the field solution. The middle column contains CPTG quantities obtained from that solution. The right column contains catalog information that is not supplied as a separate Structural Mode or CSMI variable and is retained for downstream structural comparison or observational control. Catalog distance D is a special case: it is not supplied as a separate Structural Mode variable, but the tabulated physical radii inherit the adopted SPARC distance scale, so D is retained explicitly as an observational control.

Field inputs	CPTG outputs	Withheld structure
r, $V_{\rm obs}$, σ_V	$g_{\rm sup}(r)$ and local $a_\star(r)$	Hubble type T
$V_{\rm gas}$, $V_{\rm disk}$, $V_{\rm bulge}$	κ_C, A_C, c_C, solved $g_{\rm CPTG}(r)$	$L_{3.6}$, $R_{\rm eff}$, $SB_{\rm eff}$
Measured radial support shape	C_N, w_N, λ, continuous N, CSMI	$R_{\rm disk}$, $SB_{\rm disk}$, $M_{\rm HI}$, $R_{\rm HI}$, $V_{\rm flat}$
Q and i only for sample selection	numerical closure and residual diagnostics	D and radial point count as observational controls

Table 3: Data-role firewall for the galaxy Structural Mode analysis. The right-column quantities are not supplied as separate Structural Mode or CSMI variables. Catalog distance D is retained as an observational control because the tabulated physical radii inherit the adopted SPARC distance scale.

SPARC's numerical Hubble coordinate runs from earlier disks toward later and irregular systems: $T = 0$ denotes S0, $T = 1$–4 Sa–Sbc, $T = 5$–8 Sc–Sdm, $T = 9$ Sm, $T = 10$ Im, and $T = 11$ BCD [8]. This gives a convenient external ordering for testing whether Structural Mode recovers recognizable organization without receiving that ordering as input.

3.2 Resolved CPTG gravitational field

The galaxy gravitational calculation is presented first because the Structural Mode readout is defined on the completed CPTG field. The earlier galaxy tutorial documents a historical single-scale realization [3]; the current companion calculation paper documents the resolved measured-field workflow and worked radial solutions [4].

3.2.1 Observed support and baryonic source

At each measured radius the observed rotation speed determines the acceleration required to support the rotating galaxy,

$$g_{\rm sup}(r_i) = \frac{V_{{\rm obs},i}^2}{r_i}, \qquad a_{\star,i} = g_{\rm sup}(r_i). \tag{7}$$

The local $a_\star$ is therefore a structural acceleration coordinate of the resolved state, not a universal acceleration constant. Its value at each measured radius remains distinct from the differentiable support reconstruction used to evaluate the curvature, polarization, and transport gradients. It is recalculated independently at every radius and is carried directly into both the polarization amplitude and the nonlinear response factor. In this sense $a_\star$ is one of the quantities that shapes the solved radial field from which Structural Mode is later measured. The baryonic source is constructed directly from the tabulated gas, disk, and bulge contributions,

$$V_b^2 = Q(V_{\rm gas}) + Q(V_{\rm disk}) + Q(V_{\rm bulge}), \qquad Q(V) = V|V|, \tag{8}$$

with

$$g_b(r) = \frac{V_b^2(r)}{r}. \tag{9}$$

The stellar templates are retained at their unit-Υ reference normalization; no additional fitted disk or bulge mass-to-light rescaling is introduced by this source construction.

3.2.2 Structure-dependent response of the support field

The solved response must know not only the local acceleration but also how the support field changes with radius. In the effective radial galaxy realization, the field-and-gradient carrier is

$$C_C(r) = g_{\rm sup}^2(r) + \kappa_C^2 \left\{ \left[\frac{dg_{\rm sup}}{dr} \right]^2 + 2 \left[\frac{g_{\rm sup}(r)}{r} \right]^2 \right\}. \tag{10}$$

The reduced length κ_C controls the contribution of spatial gradients. From this carrier the bare polarization response is

$$B_{\rm pol}(r) = \frac{1}{r^2} \left[(r^2 + 2\kappa_C^2) g_{\rm sup} C_C^{-2/3} - \frac{d}{dr} \left(r^2 \kappa_C^2 \frac{dg_{\rm sup}}{dr} C_C^{-2/3} \right) \right]. \tag{11}$$

A second coordinate records the local radial organization of the support field,

$$\Sigma_C^2(r) = \frac{1}{3} \left[1 - \frac{d \ln g_{\rm sup}(r)}{d \ln r} \right]. \tag{12}$$

The resolved calculation requires $\Sigma_C^2 \geq 0$, without imposing an upper bound. Where necessary, the differentiable support reconstruction is constrained to maintain nonnegativity; valid values exceeding unity remain unchanged and are not clipped. This numerical admissibility condition does not alter the measured local $a_\star$.

At high acceleration the additional response is moderated by the registered nonlinear factor

$$\mathcal{S}(r) = \left[1 + \left(\frac{g_{\rm CPTG}(r)}{a_\star(r)} \right)^{123/50} \right]^{-11/50}. \tag{13}$$

The resolved CPTG field is then

$$g_{\mathrm{CPTG}}(r) = \frac{g_b(r) + A_C a_\star^{2/3}(r) B_{\mathrm{pol}}(r) \mathcal{S}(r)}{1 + c_C \Sigma_C^2(r)}. \tag{14}$$

The object-level quantities κ_C, A_C, and c_C are reduced state/closure coordinates of that galaxy solution. They are not universal constants. Their numerical role is to close the measured support state through the polarization and transport response already defined above. Dimensional consistency gives $[\kappa_C] = L$, $[A_C] = L^{2/3}T^{-4/3}$, and $[c_C] = 1$.

3.2.3 Gravitational closure and residual diagnostics

Before asking anything about Structural Mode, the gravitational solution itself must pass. At each radial point the field equation is checked through

$$\mathcal{R}_i^{\mathrm{field}} = [1 + c_C \Sigma_{C,i}^2] g_{\mathrm{CPTG},i} - g_{b,i} - A_C a_{\star,i}^{2/3} B_{\mathrm{pol},i} \mathcal{S}_i = 0. \tag{15}$$

Each resolved point is checked against Eq. (15) after the implicit solve. Numerical equation closure is separate from the observational residual statistics reported below.

The observational residual summary over the 3168 resolved points is

$$\chi^2_{\mathrm{CPTG}} = 16472.292025, \qquad \chi^2_{\mathrm{CPTG}}/3168 = 5.199587, \tag{16}$$

The median of the galaxy-level χ^2-per-point values is 0.754405, and the pooled CPTG radial-acceleration-relation (RAR) residual scatter is 0.102497 dex. The aggregate χ^2 and the galaxy median characterize different aspects of the same measured sample; the RAR statistic describes its logarithmic acceleration residuals. These are conditioned reconstructions based on measured radial support, not source-only prospective predictions. With closure established, the Structural Mode operator is applied to the retained field.

3.3 Structural Mode in galaxies

3.3.1 The radial organization measurement

For a solved radial galaxy field $g(r)$, the outer structural scale is the outer retained radius,

$$R = \max(r_i). \tag{17}$$

The galaxy operator then defines

$$C_N(r) = g^2(r) + \kappa_{\mathrm{struct}}^2 \left(\frac{dg}{dr}\right)^2, \qquad \kappa_{\mathrm{struct}} = 1 \text{ kpc}, \tag{18}$$

with structural weight

$$w_N(r) = \left|\frac{dC_N}{dr}\right|. \tag{19}$$

The weight is large where the solved structural carrier changes rapidly. The registered interval $R/5 \le r \le R$ restricts the readout to the resolved outer structural field rather than allowing the innermost source-dominated region to set the centroid. The same fractional interval and the same κ_{struct} are used throughout the galaxy sample; neither is selected from morphology. The organization length is therefore

$$\lambda_{N,\mathrm{gal}} = \frac{\displaystyle\int_{R/5}^{R} r\, w_N(r)\, dr}{\displaystyle\int_{R/5}^{R} w_N(r)\, dr}, \qquad N_{\mathrm{gal}} = \frac{R}{\lambda_{N,\mathrm{gal}}}. \tag{20}$$

Because $R/5 \le \lambda \le R$ whenever the structural weight is nonzero, the continuous galaxy realization obeys $1 \le N \le 5$. The important point is not the numerical bound itself but the meaning of the centroid: N reports where the changing part of the solved field is concentrated. A purely multiplicative rescaling of the structural weight cancels between numerator and denominator, which is one reason the coordinate reads organization more directly than overall amplitude.

3.3.2 Numerical realization on the resolved radial grid

The SPARC radii are generally nonuniform, so the sampled estimator used for the reported galaxy values is stated separately from the continuum functional above. After sorting the retained radii, D_r is the registered nonuniform-grid gradient operator: it evaluates the derivative against the radius coordinate itself, using centered differences in the interior and one-sided endpoint differences. With that operator, define

$$C_i = g_i^2 + \kappa_{\mathrm{struct}}^2 (D_r g)_i^2, \qquad u_i = \left|(D_r C)_i\right|. \tag{21}$$

Only points satisfying $r_i > R/5$ are retained. If n_{out} is the number of retained outer points, the registered filter $\mathcal{S}_5$ is the equal-weight five-point convolution with kernel $(1,1,1,1,1)/5$, returned on the same-length outer array; the endpoint operation therefore uses the registered zero extension outside the retained array. The sampled structural weights are

$$w_i = \begin{cases} (\mathcal{S}_5 u)_i, & n_{\mathrm{out}} > 5, \\ u_i, & n_{\mathrm{out}} \le 5. \end{cases} \tag{22}$$

The numerical centroid and Structural Mode are then

$$\lambda_{\mathrm{num}} = \frac{\sum_i r_i w_i}{\sum_i w_i}, \qquad N_{\mathrm{num}} = \frac{R}{\lambda_{\mathrm{num}}}. \tag{23}$$

No explicit radial-cell factor Δr_i enters these sums. Equation (23) is therefore the registered sampled structural estimator, not a claim that the nonuniform discrete sums are a quadrature rule for Eq. (20). This distinction is why the point-count, distance, and resolution controls below are part of the validation. Hereafter, reported empirical galaxy N values refer to N_{num} unless the continuum functional is explicitly identified.

The current 153-galaxy primary sample gives

$$\langle N \rangle = 2.237438, \qquad \mathrm{median}(N) = 2.150980, \qquad 1.265930 \leq N \leq 3.772229. \tag{24}$$

3.3.3 Comparison with withheld structural observables

The cleanest first comparison is the withheld SPARC Hubble-type coordinate. Across all 153 primary galaxies,

$$\rho_S(N, T) = -0.673451, \qquad p = 1.47 \times 10^{-21}, \tag{25}$$

with a 5000-resample galaxy bootstrap 95% interval (fixed random seed 20261010) of approximately

$$-0.7596 \leq \rho_S(N, T) \leq -0.5683. \tag{26}$$

Since larger SPARC T corresponds to later and more irregular systems, the negative correlation means that larger Structural Mode is associated statistically with earlier, more centrally organized galaxy populations. By the firewall in Table 3, T is a withheld comparison coordinate.

The relation is not confined to a single morphology coordinate. Table 4 compares current N with the family of non-input SPARC structural quantities.

Comparison/control	n	$\rho_S(N, X)$	p
Effective surface brightness SB_{eff}	153	+0.68365	2.08×10^{-22}
Hubble type T	153	−0.67345	1.47×10^{-21}
3.6 μm luminosity $L_{3.6}$	153	+0.64804	1.37×10^{-19}
H I radius R_{HI}	149	+0.63464	3.61×10^{-18}
H I mass M_{HI}	153	+0.62672	4.50×10^{-18}
Flat velocity V_{flat}	123	+0.62364	1.31×10^{-14}
Disk scale length R_{disk}	153	+0.47733	4.42×10^{-10}
Disk surface brightness SB_{disk}	153	+0.43546	1.85×10^{-8}
Distance D	153	+0.33921	1.79×10^{-5}
Effective radius R_{eff}	153	+0.34355	1.38×10^{-5}

Table 4: Current monotonic relations between Structural Mode and SPARC comparison/control quantities that are not supplied as separate N/CSMI variables. Catalog distance is included as an explicit observational control because the tabulated physical radii inherit the adopted SPARC distance scale.

These correlations are expected to overlap physically: if N measures gravitational organization, it should covary with other measurements of that organization. The listed quantities are not supplied as separate N/CSMI variables. Distance D is treated specifically as an observational control because the physical radial coordinate already inherits the adopted SPARC distance scale.

3.3.4 Radial sampling and distance controls

Structural Mode is measured from a resolved radial field, so the number and placement of radial measurements matter. In the current sample, N itself correlates with radial point count, $\rho_S(N, n_{\rm pt}) = +0.68380$. The morphology relation must therefore survive an explicit sampling control before it can be interpreted structurally. The partial Spearman calculation rank-transforms the variables, regresses each target rank on the declared control ranks, and correlates the resulting residuals.

After rank-controlling for point count,

$$\rho_S(N, T \mid n_{\rm pt}) = -0.50862, \qquad p = 2.23 \times 10^{-11}, \tag{27}$$

and controlling jointly for point count and distance gives

$$\rho_S(N, T \mid n_{\rm pt}, D) = -0.38392, \qquad p = 1.14 \times 10^{-6}. \tag{28}$$

The raw relation is reduced, as it should be if observational resolution carries information, but it does not disappear. Restricting the sample instead to progressively better-resolved galaxies gives

Min. radial points	Galaxies	$\rho_S(N, T)$	p
10	115	−0.66600	4.60×10^{-16}
15	81	−0.66187	1.73×10^{-11}
20	58	−0.69143	1.87×10^{-9}

Table 5: Resolution-restricted Structural Mode ordering under the current solved fields. The relation persists as the sparsest rotation curves are removed.

3.3.5 Population ordering across Structural Mode

The statistics above become easier to visualize conceptually by dividing the 153 galaxies into quartiles of measured N. The median catalog type changes monotonically from $T = 9$ in the lowest-mode quartile to $T = 3$ in the highest-mode quartile. Median effective surface brightness rises from 26.11 to 716.48 $L_\odot\,{\rm pc}^{-2}$, while the median measured flat velocity among galaxies with available $V_{\rm flat}$ rises from 83.8 to 184.7 $\rm km\,s^{-1}$.

A still simpler broad contrast is useful. Among the 76 galaxies with $N \leq 2.15$, 44 are in the $T = 9$–11 Sm/Im/BCD family and only 5 are in the $T = 0$–4 early/intermediate family. Among the 29 galaxies with $N > 2.80$, none is irregular/BCD and 21 are in the early/intermediate family. This does not turn CSMI into Hubble morphology. It shows that a morphology-blind gravitational coordinate produces a strong population ordering that can then be compared with appearance-based classifications.

3.4 Independent structural corroboration in the primary SPARC sample

The primary-sample statistics establish the population-level Structural Mode ordering. Two galaxies make the physical meaning especially clear because their current primary-sample N/CSMI readouts indicate substantially more organized disk structure than their conventional SPARC classifications, while independent resolved observations directly corroborate that organization. These objects are presented as independently corroborated primary-sample cases: the external observations resolve the extended disk, bar, or spiral structure indicated by the post-solution Structural Mode readout.

3.4.1 NGC 3741: hidden disk and spiral organization

SPARC classifies NGC 3741 as Im, $T = 10$. Its current primary-sample CPTG solution gives

$$N = 2.214270, \qquad \text{CSMI} = \text{LSB Spiral}. \tag{29}$$

That downstream label is not imposed from morphology. Independent observations instead make the stronger solved-field organization physically intelligible. H I measurements establish an exceptionally extended rotating gas disk [9]; pattern-speed work directly identifies a slow gaseous bar and associated non-axisymmetric structure [10]; and deep LBT/HST imaging reveals a bar-like stellar feature together with central H I spiral arms and extended star formation [11]. The current Structural Mode readout therefore captures organized disk structure that the conventional Im label alone does not describe.

3.4.2 NGC 2915: extended spiral structure beneath a compact optical identity

SPARC classifies NGC 2915 as a blue compact dwarf, $T = 11$. Its current primary-sample CPTG solution gives

$$N = 2.611265, \qquad \text{CSMI} = \text{Late Spiral}. \tag{30}$$

Resolved H I observations show that the compact blue stellar component sits inside a much larger rotating disk with coherent bar and spiral structure. The foundational synthesis study described NGC 2915 as a spiral system with a blue compact dwarf core [12], and later H I work continued to resolve the extended organized disk [13]. The current CSMI label is not asserted to be a literal optical Hubble subtype; rather, the Structural Mode measurement correctly identifies substantially stronger gravitational organization than the compact optical classification alone suggests.

3.5 Robustness and interpretation

A structural coordinate is scientifically useful only if its observational interpretation remains meaningful across varied radial sampling. The current 153-galaxy solution supports two complementary checks. First, the morphology association survives explicit rank controls

for point count and catalog distance in Eqs. (27)–(28). Second, the resolution-restricted samples in Table 5 retain strong negative N–T correlations after excluding sparsely sampled galaxies.

Figure 1 displays the current individual measured N values against the withheld SPARC Hubble-type coordinate. It uses the same 153 primary-sample solutions and catalog metadata as the tables above. The population ordering is evident while the spread at each type also shows that N is not a deterministic morphological label.

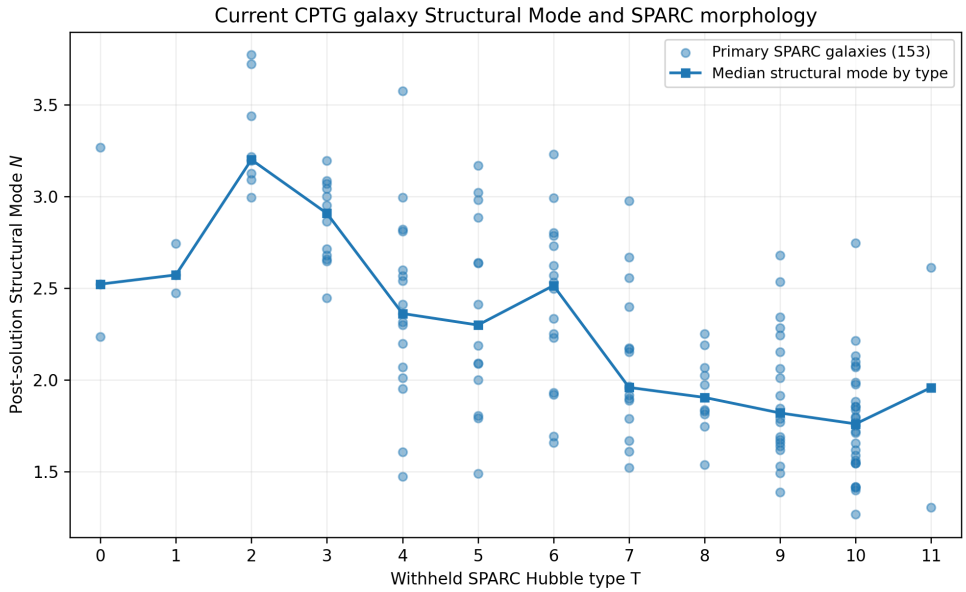


Figure 1: Current primary-sample galaxy Structural Mode versus withheld SPARC Hubble type for 153 galaxies. Each point represents the post-solution N fingerprint of one galaxy; the overlaid line joins the median N at each populated Hubble type. The Spearman association is $\rho_S = -0.673451$ ($p = 1.47 \times 10^{-21}$). Morphology is not an input to the CPTG solve, the Structural Mode operator, or CSMI assignment.

These results define the correct claim boundary. Structural Mode is not a prospective forecast of a galaxy's morphology. It is measured after the gravitational solution exists. The primary-sample finding is that a morphology-blind gravitational measurement produces a coherent ordering across withheld SPARC structural coordinates while surviving explicit sampling and resolution controls. A future independent survey can make that use prospective by preregistering the field equations, the N operator, and the specific CSMI catalog revision used for that test before the new catalog is examined. Afterward, a larger solved population may justify a CSMI revision without altering the earlier measurements.

4 Galaxy Analysis with the CPTG SPARC Browser Workbench

The CPTG SPARC Browser Workbench v1.13.1 is an interactive application for examining galaxy rotation curves. For individual galaxies or primary-sample batches, it implements the resolved CPTG procedure documented in the companion galaxy calculation paper [4] using published SPARC measurements and baryonic source components.

For each galaxy, the interface displays the signed baryonic source, measured local structural acceleration, differentiated support, polarization, and transport alongside the fitted coefficients and solved velocities. Post-solution Structural Mode N, structural scale λ_{num}, and CSMI classification are reported separately, allowing the gravitational reconstruction to be examined at each radius.

The browser supports galaxy search, batch runs, rotation-curve plots, and numerical exports. SPARC velocities are displayed beside CPTG solutions and the fixed-a_0 MOND reference, with separate velocity χ^2 diagnostics.

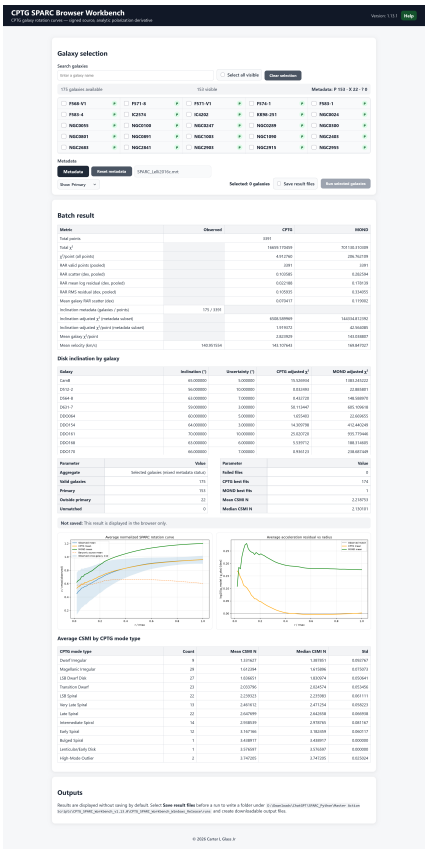


Figure 2: The CPTG SPARC Browser Workbench, showing batch-analysis results, sample statistics, and structural measurements.

The Radial Acceleration Relation (RAR) diagnostics are evaluated separately from velocity-based χ^2. At each valid radius, the observed acceleration $g_{\mathrm{obs}} = V_{\mathrm{obs}}^2/r$ is compared with each calculated acceleration. The logarithmic residual $\log_{10}(g_{\mathrm{model}}/g_{\mathrm{obs}})$ supplies the RAR scatter in dex, mean residual, RMS residual, and valid-point count for CPTG and MOND independently. These acceleration-space measurements do not alter either gravitational calculation.

Official SPARC metadata provide galaxy classification, distance, disk inclination, inclination uncertainty, and sample membership. Where available, inclination-adjusted velocity statistics are displayed separately without modifying the standard CPTG solution.

Batch summaries retain individual galaxy results while reporting successful and failed counts, primary-sample coverage, aggregate velocity χ^2, pooled RAR statistics, mean galaxy RAR scatter, and Structural Mode measurements. Full-precision radius-level exports, rotation-curve figures, and downloadable result packages support reproduction of the worked examples and further SPARC analysis [2].

5 Galaxy Clusters

5.1 Pressure-supported state and structural acceleration

Galaxy clusters are pressure-supported rather than rotation-supported, so their source state is reconstructed hydrostatically or through the Jeans equation. The resolved cluster realization evaluates

$$a_\star(r_i) = g_{\rm sup}(r_i) \tag{31}$$

at every retained radius. This local scale enters the pressure-supported CPTG response through the polarization weighting and nonlinear response factor; the resulting spherical field is then passed to the cluster Structural Mode operator.

5.2 Spherical organization functional

A galaxy cluster cannot be treated as a scaled-up SPARC disk. For a pressure-supported spherical cluster, Structural Mode is rederived from the cluster's solved organization scale. For solved cluster field $g_{\rm CPTG}(r)$, define [6, 14, 15]

$$C_{N,C}(r) = 3g^2_{\rm CPTG}(r) + \left[r\frac{dg_{\rm CPTG}}{dr}\right]^2, \tag{32}$$

and

$$w_{N,C}(r) = \left|\frac{dC_{N,C}}{dr}\right|. \tag{33}$$

For each reconstruction, R is the outer retained radius of that solved cluster field. The $R/5 \le r \le R$ interval is applied within that reconstruction, so independent X-COP and ACCEPT solutions retain their own structural domains rather than being forced onto a common averaged profile. The three-dimensional shell measure then gives

$$\lambda_{N,C} = \frac{\displaystyle\int_{R/5}^{R} r^3 w_{N,C}(r)\,dr}{\displaystyle\int_{R/5}^{R} r^2 w_{N,C}(r)\,dr}, \qquad N_C = \frac{R}{\lambda_{N,C}}. \tag{34}$$

The factor of three and the shell weighting arise from the spherical organization of the radial field. The cluster realization therefore preserves the common meaning $N = R/\lambda$ without copying the galaxy 1-kpc gradient operator into a three-dimensional pressure-supported system.

5.3 Seven-cluster measurements

The current seven-cluster program carries two independently reconstructed support fields. X-COP supplies the primary Structural Mode measurement; ACCEPT supplies a separate reconstruction-sensitivity measurement [16, 6]. For compact table notation only, $\lambda_C \equiv \lambda_{N,C}$. The resulting values are

Cluster	X-COP N_C	X-COP λ_C (kpc)	ACCEPT N_C	ACCEPT λ_C (kpc)
A2029	1.521388	424.829	1.397476	349.863
A1795	1.566115	342.628	1.507837	245.288
A2142	1.328084	486.664	1.760504	220.087
A2319	1.350667	456.777	1.400538	333.231
A644	1.400836	401.293	1.120615	436.301
A85	1.533717	366.525	1.428954	162.552
ZW1215	1.358032	454.300	1.304734	430.850

Table 6: Post-solution spherical Structural Mode for the seven cluster fields. Independent X-COP and ACCEPT reconstructions remain separate measurements and are never averaged.

The primary X-COP values span $N_C = 1.328084$–1.566115. The ACCEPT values do not have to match them exactly because Structural Mode depends on derivatives of the solved field, the retained radial domain, and the reconstruction of the pressure-support state. The difference between two valid reconstructions is itself structural information about the sensitivity of the inferred organization.

A cluster CSMI is therefore possible in principle, but seven objects are not used here to manufacture one. A useful cluster taxonomy would require a broader population of solved pressure-supported systems and a physically interpretable set of recurring mode populations or transitions.

6 Black Holes

6.1 Compact structural state

The compact realization uses the strong-field acceleration normalization g_g from Eq. (5) together with the solved spin state. With

$$x = \frac{r}{r_m}, \qquad g(r) = g_g U(x), \tag{35}$$

mass fixes the physical acceleration and length scales while $U(x)$ and $\hat{a}_\infty$ determine the dimensionless finite geometry. Structural Mode is evaluated only after that compact mass–spin state has closed [5, 6].

6.2 Relativistic coordinate realization

The relativistic compact geometry requires its own organization measure rather than the resolved galaxy or cluster operator.

For the Kerr-compatible finite mass-function geometry, define

$$\Sigma = r^2 + a^2 \cos^2 \theta, \qquad a = \hat{a}_\infty r_m, \tag{36}$$

with exact determinant measure

$$\sqrt{-g} = \Sigma \sin \theta. \tag{37}$$

The determinant-based stationary radial–angular measure accumulated to radius r is

$$\mathcal{V}(r) = \frac{4\pi}{3} r(r^2 + a^2), \tag{38}$$

which defines an equal-measure shell radius

$$\varrho(r, a) = \left[r(r^2 + a^2) \right]^{1/3}. \tag{39}$$

After normalization by the compact matching scale, the rotating organization coordinate is

$$\xi(x, \hat{a}_\infty) = \left[\frac{x(x^2 + \hat{a}_\infty^2)}{1 + \hat{a}_\infty^2} \right]^{1/3}, \qquad x = \frac{r}{r_m}. \tag{40}$$

6.3 Finite regular profile, structural weight, and mode

The compact structural operator is built from the same finite mass-function realization that defines the interior geometry. A finite spherical center permits $m(r) = O(r^3)$, while the retained rotating Kerr-compatible continuation uses the minimum analytic odd behavior $m(r) = O(r^5)$ required by its regular-center construction [5]. Writing

$$m(r) = M_\infty F(x), \qquad x = \frac{r}{r_m}, \qquad 0 \le x \le 1, \tag{41}$$

and requiring smooth matching to the constant-mass exterior through second radial derivative,

$$F(1) = 1, \qquad F'(1) = 0, \qquad F''(1) = 0, \tag{42}$$

within the retained minimum-degree analytic odd-polynomial realization uniquely gives the profile

$$F(x) = \frac{1}{8} \left(63x^5 - 90x^7 + 35x^9 \right). \tag{43}$$

No additional compact length or object-specific interior shape coefficient is introduced. The corresponding dimensionless acceleration profile is

$$U(x) = \frac{F(x)}{x^2} = \frac{1}{8}\left(63x^3 - 90x^5 + 35x^7\right), \tag{44}$$

so the strong-field normalization g_g supplies the physical acceleration scale while $U(x)$ supplies the finite radial shape. The Structural Mode construction then uses the invariant

$$\mathcal{C}(x) = U(x)^2 + U'(x)^2 + 2\left[\frac{U(x)}{x}\right]^2, \qquad w(x) = |\mathcal{C}'(x)|\,. \tag{45}$$

The equal-measure coordinate converts the weighted dimensionless centroid into the physical compact organization length

$$\lambda_{N,\mathrm{BH}} = r_m \frac{\int_0^1 \xi(x, \hat{a}_\infty) w(x)\, dx}{\int_0^1 w(x)\, dx}. \tag{46}$$

The rotating compact Structural Mode therefore retains the same R/λ meaning used in the other regimes,

$$N_{\mathrm{BH}}(\hat{a}_\infty) = \frac{r_m}{\lambda_{N,\mathrm{BH}}} = \frac{\int_0^1 w(x)\, dx}{\int_0^1 \xi(x, \hat{a}_\infty) w(x)\, dx}. \tag{47}$$

Mass, r_m, and c cancel from the dimensionless ratio in Eq. (47). The resulting fingerprint is controlled by the solved exterior spin magnitude in the registered rotating realization. The spherical limit is

$$N_{\mathrm{BH}}(0) = 1.912233171442, \tag{48}$$

and N_{BH} decreases monotonically with spin magnitude toward the high-spin end of the registered domain [6].

The Structural-Mode organization shell is recovered from the solved mode through

$$x_N\left(x_N^2 + q^2\right) = \frac{1 + q^2}{N_{\mathrm{BH}}^3}, \qquad r_N = x_N r_m, \qquad q = |\hat{a}_\infty|. \tag{49}$$

Thus N_{BH} is not merely a scalar label. It identifies a definite shell inside the finite compact geometry.

6.4 Two benchmark black holes

Two deliberately contrasting stellar-mass black holes provide compact Structural Mode benchmarks under the same mass–spin relation and rotating registry [6, 17, 18, 19, 20, 21, 22]. Their central branches are

Object	$M_\infty/M_\odot$	$\lvert\hat{a}_\infty\rvert$	r_m (km)	N_{BH}	Structure
M33 X-7	15.65	0.84	23.1099	1.49387	High-spin, lower mode
LMC X-3	6.98	0.21	10.3072	1.83197	Low-spin, higher mode

Table 7: Two compact Structural Mode benchmarks evaluated with the same registered mass–spin and equal-measure construction.

Together the two benchmarks show how the same compact registry responds at widely separated spin states. The GWTC calculation then applies that unchanged construction across an entire binary-black-hole catalog.

6.5 GWTC-5.0 population: 520 compact components

6.5.1 Catalog-wide compact-object extension

The GWTC-5.0 compact-state analysis applies one fixed CPTG compact construction to the declared 259-event binary-black-hole subset retained for this analysis [23, 24, 25, 7]. This is the CPTG compact-state analysis set rather than a claim about the size of the full cumulative compact-binary catalog. The retained registry contains 260 unique numerical streams and therefore 520 compact components. Structural Mode is evaluated only after each mass–spin state has closed.

Across the 520 retained component medians,

$$1.473148 \leq N_{\mathrm{BH}} \leq 1.898389. \tag{50}$$

No catalog-average Structural Mode is assigned. Every value remains attached to its own component, posterior reconstruction, exterior mass scale, and spin state.

The full posterior audit also gives the radial ordering

$$x_N < x_- < 1, \tag{51}$$

where x_- is the inner trapping-surface coordinate and the compact matching boundary is $x = 1$. The minimum recovered sample-wise separation is

$$\min(x_- - x_N) = 0.049408417286901. \tag{52}$$

The Structural-Mode organization shell is therefore geometrically distinct from both the trapping surface and the exterior handoff throughout the retained catalog. Figure 3 shows that normalized architecture across the registered spin domain together with the retained GWTC component medians.

The catalog result extends the same compact Structural Mode construction across the full declared population without changing the operator from object to object.

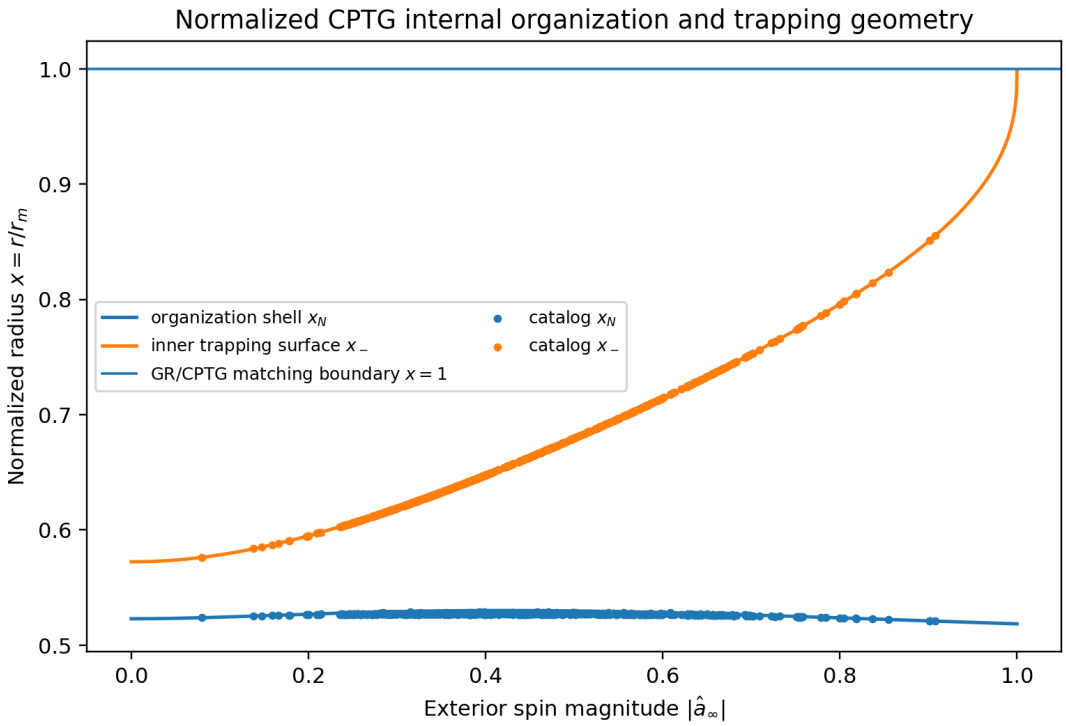


Figure 3: Normalized compact organization and trapping geometry. The Structural-Mode shell x_N remains inside the inner trapping surface x_-, while the GR/CPTG matching boundary is $x = 1$. Points show the retained GWTC compact-component medians on the same registered curves.

6.6 Structural Mode and the local structure of time

6.6.1 Organization shell and local clock rate

The compact realization adds a physical possibility beyond a purely spatial organization measurement. Once Structural Mode has fixed r_N, the finite geometry supplies a local lapse field across that structurally selected shell. Structural Mode does not replace the lapse field; it provides the structural address at which the determinant-weighted angular clock readout is evaluated.

Using the same finite profile $F(x)$ from Eq. (43) and the spin magnitude $q = |\hat{a}_\infty|$, define the dimensionless rotating metric functions

$$\delta(x, q) = x^2 - 2xF(x) + q^2, \tag{53}$$

$$\sigma(x, \theta, q) = x^2 + q^2 \cos^2 \theta, \tag{54}$$

$$\mathcal{A}_d(x, \theta, q) = (x^2 + q^2)^2 - q^2 \delta(x, q) \sin^2 \theta. \tag{55}$$

The physical inner trapping surface is the root $0 < x_- < 1$ of

$$\delta(x_-, q) = 0, \qquad r_-^{\mathrm{CPTG}} = x_- r_m. \tag{56}$$

For the locally nonrotating, or ZAMO, geometry, the lapse is

$$\alpha(x,\theta;q)=\sqrt{\frac{\delta\,\sigma}{\mathcal{A}_d}}, \tag{57}$$

and the rotating shell readout uses the determinant-measure angular average

$$\bar{\alpha}(x;q)=\frac{\displaystyle\int_0^{\pi}\alpha(x,\theta;q)\,\sigma(x,\theta,q)\sin\theta\,d\theta}{\displaystyle\int_0^{\pi}\sigma(x,\theta,q)\sin\theta\,d\theta}. \tag{58}$$

At the Structural-Mode shell,

$$T_{24,N}(q)=24\,\bar{\alpha}(x_N(q);q)\ \text{ h}. \tag{59}$$

Both x_N and $\bar{\alpha}_N$ are dimensionless functions of the solved spin state, so the Structural-N clock is scale-free within the registered compact realization. Over the full mathematical spin interval,

$$T_{24,N}(0)=11.592884\text{ h},\qquad T_{24,N}\to 20.461495\text{ h}\quad(q\to 1^{-}). \tag{60}$$

Across the 520 retained GWTC component medians, the derived range is

$$11.852622\le T_{24,N}\le 19.985616\text{ h} \tag{61}$$

per 24 source-local asymptotic exterior Kerr hours [7].

The compact result extends Structural Mode beyond spatial organization: the measured organization shell carries a definite determinant-weighted angularly averaged time-dilation readout. In this sense N reveals the structure of time by locating the shell at which the solved geometry supplies that clock readout.

6.6.2 Near-trapping-surface clock scaling

The Structural-N clock should be distinguished from the local clock obtained arbitrarily close to the inner trapping surface. With

$$d=r_{-}^{\mathrm{CPTG}}-r>0,\qquad \epsilon=\frac{d}{r_m}, \tag{62}$$

where d is a Boyer–Lindquist radial-coordinate offset, the simple-root expansion of δ and the determinant-weighted lapse give

$$T_{24}(d)\simeq 24\,\mathcal{C}(q)\sqrt{\frac{d}{r_m}}\ \text{h}, \tag{63}$$

where $\mathcal{C}(q)$ is the spin-dependent coefficient derived in the compact clock calculation [7]. Fits to the GWTC 1 cm, 10 cm, and 1 m component-median clock readouts independently

recover

$$0.49999464 \leq \beta \leq 0.50001636, \qquad \text{median}(\beta) = 0.50000089, \tag{64}$$

for $T_{24} \propto d^{\beta}$. The near-boundary clock is scale-dependent, whereas the Structural-N clock is evaluated at the organization shell fixed by the mode; the two are physically distinct.

6.7 Structural Mode and geometric rotation

The same organization shell also provides a natural location at which to evaluate the angular velocity of the local zero-angular-momentum frame. In the retained finite geometry, the magnitude of the geometric frame-dragging rate is

$$\omega(x, \theta; q) = \frac{c}{r_m} \frac{2q\, xF(x)}{\mathcal{A}_d(x, \theta, q)}. \tag{65}$$

This is a spacetime frame-dragging rate relative to source-local asymptotic exterior Kerr coordinate time, not a rigid-body or material rotation law. Its determinant-measure angular average is

$$\bar{\omega}(x; q) = \frac{\displaystyle\int_0^{\pi} \omega(x, \theta; q)\, \sigma(x, \theta, q) \sin\theta \, d\theta}{\displaystyle\int_0^{\pi} \sigma(x, \theta, q) \sin\theta \, d\theta}. \tag{66}$$

Equivalently, define the dimensionless rate $\hat{\omega} = (r_m/c)\omega$. At the Structural-Mode shell,

$$\bar{\omega}_N = \frac{c}{r_m} \hat{\omega}_N(q), \tag{67}$$

so the solved spin state fixes the dimensionless rotational geometry while the compact matching radius restores the physical angular-rate scale.

Across the retained GWTC component medians, the determinant-averaged Structural-N frame-dragging rate spans

$$389.688 \leq \bar{\omega}_N \leq 16330.834 \text{ rad}\,\text{s}^{-1}. \tag{68}$$

Together with Eq. (61), these rates show that the Structural-Mode shell carries a local spacetime environment: an organization coordinate, determinant-weighted angular averages of the proper-time rate relative to the source-local exterior clock, and the geometric frame-dragging rate.

7 Cross-Regime Interpretation

The three realizations differ in geometry but share one measurement principle: each regime supplies its own organization length λ, and the continuous coordinate $N = R/\lambda$ locates the solved organization relative to the declared structural extent. Numerical

equality among galaxy, cluster, and compact values is neither required nor expected, and no downstream catalog label is transferred between object classes.

Structural Mode is a measurement rather than a prospective prediction of an unsolved object. Its scientific value lies in relationships exposed after the solve: the primary SPARC analysis recovers a strong morphology-blind ordering across withheld structural coordinates; independent cluster reconstructions retain distinct fingerprints; and the compact population produces a well-ordered organization shell with definite clock and frame-dragging readouts. Statistical models may use measured N values predictively, but that is downstream prediction from an already measured structural coordinate.

Finally, N should not be interpreted as a measure of total gravitational strength. It records where the solved organization is concentrated. In the compact realization mass cancels from the dimensionless $N_{\rm BH}$ relation and re-enters only when the organization shell is converted to a physical radius; in the galaxy sample the mode remains highly stable across a material refinement of the preceding gravitational field. Both results separate structural organization from overall response amplitude.

8 Higher Structural Resolution Beyond Structural Mode

Structural Mode intentionally compresses a solved organization profile into one global coordinate. Two objects can therefore share the same mode while differing in the width, asymmetry, multimodality, or angular structure of their response. Any higher structural coordinate should be derived from that same solved state rather than introduced as a new fitting variable.

8.1 Structural Mode as the first organization moment

Let ξ denote the dimensionless organization coordinate appropriate to the declared geometry and $d\mu$ its geometric measure. For galaxies and spherical clusters, $\xi = r/R$; for the rotating compact realization, ξ is the equal-measure coordinate of Eq. (40). If $W \geq 0$ is the structural weight, define

$$dP = \frac{W\, d\mu}{\int W\, d\mu}, \qquad \int dP = 1, \tag{69}$$

so that

$$\langle \xi \rangle = \int \xi\, dP, \qquad N = \frac{1}{\langle \xi \rangle} = \frac{R}{\lambda}. \tag{70}$$

Thus N retains the centroid of the solved organization distribution but not its shape. A natural next coordinate is the structural width

$$\sigma_\xi^2 = \int (\xi - \langle \xi \rangle)^2\, dP, \tag{71}$$

which would distinguish a tightly localized organization region from a broadly distributed response at fixed N. Higher moments could likewise characterize asymmetry or tail structure, although low-order moments alone would not uniquely resolve a genuinely multimodal profile.

8.2 Geometry-specific extensions

For galaxies, the present mode is radial and cannot retain intrinsically azimuthal information such as bars, spiral multiplicity, lopsidedness, or pitch structure. A future two-dimensional structural weight could be decomposed schematically as

$$W(r,\phi) = \sum_m W_m(r) e^{im\phi}. \tag{72}$$

The $m = 0$ component would recover the axisymmetric organization, while nonzero modes would encode departures from axisymmetry. An $m = 2$ component alone would not distinguish a bar from a two-armed spiral; that interpretation would require the radial amplitude and phase structure and an independently derived connection to resolved observations. The independently corroborated primary-sample cases NGC 3741 and NGC 2915 make the motivation concrete: their scalar N values capture stronger radial organization, while resolved observations reveal bar and spiral structure that a one-dimensional centroid intentionally discards.

For galaxy clusters, a future non-spherical structural weight could follow the three-dimensional geometry through

$$W(r,\theta,\phi) = \sum_{\ell,m} W_{\ell m}(r) Y_{\ell m}(\theta,\phi), \tag{73}$$

separating the global spherical organization from coherent ellipticity, substructure, or merger geometry. The present N_C remains the first spherical coordinate; no cluster multipole Structural Mode is asserted here.

For rotating compact objects, the determinant-averaged Structural-N shell currently compresses angular dependence in both the finite geometry and its local clock/rotation environment. A prospective axisymmetric expansion,

$$W(\xi,\theta) = \sum_\ell W_\ell(\xi) P_\ell(\cos\theta), \tag{74}$$

could retain latitudinal organization that is averaged away in the present scalar readout. Because the compact mode already selects a shell with definite lapse and frame dragging, such a spectrum could test whether the local organization of time and rotation has resolvable internal structure across the finite geometry. That possibility is prospective, not an established CPTG law.

Any higher-order structural coordinate must preserve the evidentiary ordering used here: it must be derived from the solved CPTG state, use the geometry-appropriate measure, distinguish its continuum definition from any sampled-data estimator, and

pass resolution, sampling, and uncertainty tests before receiving a catalog interpretation. Higher coordinates would add resolution to N rather than replace or retroactively alter it.

9 Falsifiability and Next Tests

The cross-regime Structural Mode claim creates direct tests that do not depend on reusing one object class to define another.

1. Future SPARC-quality galaxy samples should reproduce the morphology-blind Structural Mode ordering and the principal withheld-structure relations when the registered galaxy N operator is applied without retuning.
2. Continued refinements of the galaxy gravitational solve should leave the downstream Structural Mode population substantially stable if N is genuinely measuring organization rather than a particular response amplitude.
3. Independent cluster reconstructions with comparable radial coverage should produce interpretable, reconstruction-sensitive N_C values without averaging or retuning independent solved fields.
4. A larger pressure-supported sample should determine whether recurring cluster mode populations exist and whether a cluster CSMI is justified.
5. Future compact-object analyses should continue to satisfy the registered organization-shell ordering and reproduce the Structural-N clock and frame-dragging relations without object-specific structural coefficients.
6. A new object class should not receive Structural Mode by copying one of the existing operators. Its organization length must be derived from the geometry and solved state of that regime while preserving the common meaning of Eq. (1).
7. Any new object-class CSMI must be constructed downstream from solved Structural Mode populations and validated against independent structural information rather than imposed as an input taxonomy.

These tests make the universal claim precise. Structural Mode is general only if its physical meaning survives while the appropriate geometric realization changes.

10 Conclusion

CPTG separates the construction of a gravitational state from the measurement of its organization. The object-dependent structural acceleration scale or regime-equivalent participates in the former; Structural Mode N measures the latter through the common coordinate $N = R/\lambda$.

The galaxy program supplies the largest resolved test. Across 153 primary SPARC galaxies, current mode values span 1.265930–3.772229 and correlate strongly with the withheld Hubble coordinate, $\rho_S = -0.673451$ with $p = 1.47 \times 10^{-21}$. The relation survives explicit point-count and distance controls and persists in better-resolved subsets. The same primary sample also shows strong monotonic relations between N and withheld surface-brightness, luminosity, H I, flat-velocity, and size coordinates, while none of those quantities enters the calculation of N or CSMI. NGC 3741 and NGC 2915 provide two independently corroborated primary-sample cases in which the current morphology-blind Structural Mode readout points to stronger disk organization that is independently resolved as extended rotating, bar, and spiral structure.

The same measurement principle extends beyond galaxies without copying the galaxy operator. Pressure-supported clusters use a spherical shell measure and retain separate X-COP and ACCEPT fingerprints. Rotating finite compact objects use the determinant-based equal-measure construction; across the 259-event GWTC-5.0 analysis, 520 component fingerprints satisfy the retained structural ordering, and the resulting organization shell carries definite determinant-weighted angular averages of lapse and frame dragging. In the compact realization, Structural Mode therefore supplies the location at which the local organization of time and rotation can be read.

The continuous N value remains the fundamental measurement. CSMI is a downstream, population-dependent catalog layer whose object-class definitions may expand, subdivide, merge, or otherwise refine as larger solved populations become available. The geometry can change from galaxies to clusters to black holes, but the measurement principle remains the same: CPTG solves the gravitational state first and Structural Mode then measures how that state is organized.

Data, Code, and Evidence Availability

The numerical results used in this paper are distributed through the current CPTG evidence records. The earlier resolved SPARC analyses, cluster reconstructions, cluster Structural Mode measurements, and M33 X-7/LMC X-3 compact benchmarks are preserved with the cross-regime gravitational-law evidence archive,

`CPTG_ONE_GRAVITATIONAL_LAW_RECONSTRUCTION_EVIDENCE_20260927_FINAL.zip`

The galaxy results in this paper are restricted to the 153-object primary SPARC science sample and the associated 3168 resolved radial points. The primary-sample field solution, post-solution Structural Mode values, withheld-coordinate comparisons, and sampling controls reported here use the current catalog results and can be reproduced with the cited CPTG SPARC Browser Workbench implementation and galaxy calculation paper [4]. The compact 259-event Structural Mode, organization-shell, finite-core clock, and geometric-rotation calculations are preserved in

`CPTG_GWTC5_COMPACT_OBJECT_POPULATION_PUBLICATION_EVIDENCE_FINAL.zip`

The public CPTG repository is available at: `https://github.com/CLG2025/CPTG`.

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